

Chetan Reddy Narayanaswamy

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Education

Stanford University, CA

Sept 2024 - Jun 2026

MS in Mechanical Engineering (Robotics) | GPA: 4.0/4.0

Courses: Deep Learning for Computer Vision | Convex Optimisation | Robot Autonomy | Optimal Control | Mechatronics

Teaching Assistant, Computer Science Dept: Decision Making Under Uncertainty, Engineering Design Optimisation

Indian Institute of Technology Madras, India

Aug 2019 - Jun 2024

Masters (M.Tech) in Machine Learning, Bachelors (B.Tech) in MechE | CGPA: 9.48/10

Courses: Data Structures and Algorithms | Machine Learning | Reinforcement Learning | Kinematics and Dynamics of Machinery

KTH Royal Institute of Technology, Sweden

Aug 2022 - Jan 2023

Semester Exchange (Robotics) | School of Computer Science | GPA: 5.0/5.0

Courses: Safe Robot Planning and Control | Advanced Deep Learning | Introduction to Robotics | Sustainable Development

Publications and Patents

- **Open-H-Embodiment:** A Large-Scale Dataset for Enabling Foundation Models in Medical Robotics. Nigel Nelson, Juo-Tung Chen, Jesse Haworth, ..., **Chetan Reddy Narayanaswamy**, et al. [Link](#)
- **Clustering Users According to Causal Relationships Among User Data.** Vibhor Porwal, Harshita Chopra, Atanu R. Sinha, Sharda Kriti Modanwal, **Chetan Reddy Narayanaswamy**, Zainul Niaz. *US20250299215A1*, 2025. [Link](#)

Skills

- Robotics: Perception (3D Computer Vision), Motion Planning (A*, RRT, LQR, Optimal Control), SLAM (Visual, LiDAR)
- ML/AI: Deep Learning for Computer Vision, Reinforcement Learning, Imitation Learning, Bayesian Statistics
- Programming: Python, C++, PyTorch, OpenCV | Other Tools: ROS/ROS2, Isaac Sim, PyBullet, Gazebo, RViz, Git, Linux, W&B

Professional and Research Experience

NVIDIA x Stanford Open-H Embodiment Robotics Initiative

Stanford, CA

Open Source Dataset Generation for Healthcare Robotics Foundation Models | Skills: Python, C++, ROS

Sept 2025 - Jan 2026

- Helped develop and validate the official dataset standardization guidelines in collaboration with labs in NVIDIA, JHU and TUM
- **Engineered an automated data collection framework** in LeRobot format, contributing to open-source robot learning
- Collecting and curating 2,000+ high quality robotic surgery trajectories (needle passing, tissue retraction, suturing)

Machine Learning Intern | MORE Inc [Link](#)

San Mateo, CA

3D Computer Vision for Robotic Bin Picking | Skills: Pytorch, CI/CD, Git, Plotly, W&B

Jul 2025 - Sep 2025

- Developed **custom PyTorch operators** with learnable parameters to process 3D data optimised for real-time robotic tasks, along with an **interactive visualization dashboard** for hyperparameter tuning and model evaluation.
- Reduced model size by 99% compared to conventional CNNs on in-house 3D datasets using the geometry-aware design.

Robot Learning Researcher | CHARM and IPRL Labs, Stanford University [Link](#)

Stanford, CA

Advisors: Prof. Allison Okamura and Prof. Jeannette Bohg | Skills: C++, Isaac Sim, ROS, Imitation Learning

Jan 2025 - Jun 2025

- Built the **three-arm teleoperation** framework for the synchronised **manipulation** of the da Vinci Surgical Robot arms
- Integrated the robot SDK with ROS drivers and developed ROS Packages for seamless communication and data collection
- Trained and integrated an **ACT (Action Chunking with Transformer)** model reducing collaborative task time by 55%

Projects

Learning-Based Pick and Place Manipulation with a UR5 Robot [Link](#)

Stanford, CA

Robot Perception, Prof. Shuran Song | Skills: PyBullet, PyTorch, SE(3) Transformations

Sept 2025 - Dec 2025

- Built a UR5 manipulation pipeline in PyBullet with RGB-D sensing and TSDF-based 3D reconstruction for scene understanding
- Trained a U-Net for instance segmentation, extracted its 3D point cloud and performed 6-DoF pose estimation using ICP
- Implemented action affordance based, model free grasping for unseen objects, achieving **12/15 successful grasps**

Mapping and Object Identification for Search and Rescue with UGV [Link](#)

Geneva, Switzerland

ETH Robotics Summer School Challenge | Skills: Path Planning, SLAM, Object Detection, ROS2, Docker

Jun 2025

- Autonomously generated a 3D point cloud map and located key objects in a **30m x 30m collapsed building in 10 minutes** using a rough terrain mobile robot at a Swiss military site with realistic search and rescue environments.

- Integrated different ROS2 packages for state estimation, SLAM, ICP pose estimation, frontier exploration, and object detection

Real Time Video Segmentation for Autonomous Robotic Manipulation [Link](#)

Stanford, CA

Deep Learning with Computer Vision, Prof. Fei Fei Li | Skills: PyTorch, OpenCV, Real Time Inference

May 2025 - Jun 2025

- Used SAM2 to generate high-quality offline segmentation masks on a curated dataset and trained a lightweight U-Net on it
- Achieved 30Hz real-time inference with a **Dice score of 95.6% (60x faster compared to SAM2)** thus supporting robotic tasks